

Observability

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Observability

Definition (Observability): A linear system is **observable** if for every $T > 0$ it is possible to determine the state of the system $x(T)$ through the measurements of $y(t)$ and $u(t)$ on the interval $[0, T]$.

Recall the solution to the differential equation:

$$y(t) = Ce^{At}x(0)$$

Since we know $u(t)$, we only need to consider zero input case.

So, if $x(0)$ can be determined, then we can reconstruct $x(t)$ exactly.

Observability

Definition: A **state** $x^* \neq 0$ is said to be **unobservable** if the zero-input solution $y(t) = Ce^{At}x(0)$, with $x(0) = x^*$, is zero for all $t \geq 0$.

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So, if we can find a state x^* for which $Ce^{At}x(0) = 0$ for all $t \geq 0$, then the system is unobservable. For this to hold, all derivatives must be zero at $t = 0$.

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$$\begin{aligned}
 Ce^{At}x^* \Big|_{t=0} &= 0 \quad \Rightarrow \quad Cx^*=0 \\
 \frac{d}{dt}Ce^{At}x^* \Big|_{t=0} &= 0 \quad \Rightarrow \quad CAe^{At}x^* \Big|_{t=0} = CAx^*=0 \\
 \frac{d^2}{dt^2}Ce^{At}x^* \Big|_{t=0} &= 0 \quad \Rightarrow \quad CA^2e^{At}x^* \Big|_{t=0} = CA^2x^*=0 \\
 &\vdots \\
 \frac{d^{n-1}}{dt^{n-1}}Ce^{At}x^* \Big|_{t=0} &= 0 \quad \Rightarrow \quad CA^{n-1}e^{At}x^* \Big|_{t=0} = CA^{n-1}x^*=0
 \end{aligned}$$

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 &\vdots \\
 \frac{d^{n-1}}{dt^{n-1}}Ce^{At}x^* \Big|_{t=0} = 0 &\Rightarrow CA^{n-1}e^{At}x^* \Big|_{t=0} = CA^{n-1}x^* = 0
 \end{aligned} \right\} \underbrace{\begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}}_{W_o} x^* = 0$$

Observability

Theorem (Observability rank condition): *A linear system is observable if and only if the observability matrix reachability W_o*

$$W_o = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

has full row rank.

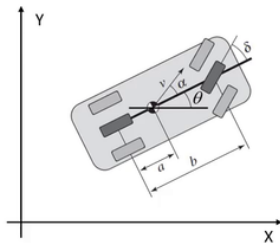
Revisit Example - Vehicle steering (Ex 7.4)

Consider the following system:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 12 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

where x_1 is the lateral position Y , x_2 is the heading orientation θ and u is the steering angle δ .



Vehicle data: $v_0 = 12 \text{ m/s}$

$a = 2 \text{ m}$

$b = 4 \text{ m}$

Example - Vehicle steering (Ex 7.4)

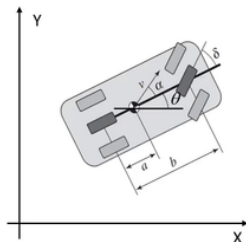
Return to our vehicle steering example, with lateral position as output signal:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 12 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} u$$

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0] u$$

Observability matrix:

$$W_o = \begin{bmatrix} C \\ CA \end{bmatrix}$$



Example - Vehicle steering (Ex 7.4)

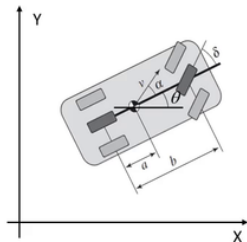
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$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0] u$$

Observability matrix:

$$W_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 12 \end{bmatrix}$$



Example - Vehicle steering (Ex 7.4)

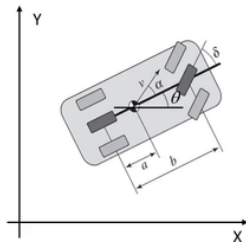
Return to our vehicle steering example, with lateral position as output signal:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 12 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} u$$

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0] u$$

Observability matrix:

$$W_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 12 \end{bmatrix}$$



Note: A square matrix M ($n \times n$) has full rank n iff the $\det(M) \neq 0$

Example - Vehicle steering (Ex 7.4)

Return to our vehicle steering example, with lateral position as output signal:

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$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + [0] u$$

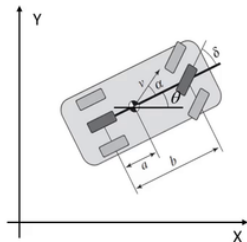
Observability matrix:

$$W_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 12 \end{bmatrix}$$

Compute the determinant:

$$\det(W_o) = \begin{vmatrix} 1 & 0 \\ 0 & 12 \end{vmatrix} = 1 \cdot 12 - 0 \cdot 0 = 12$$

The system is observable!



Note: A square matrix M ($n \times n$) has full rank n iff the $\det(M) \neq 0$

Example - Vehicle steering (Ex 7.4)

Return to our vehicle steering example, with heading angle as output signal:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 12 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

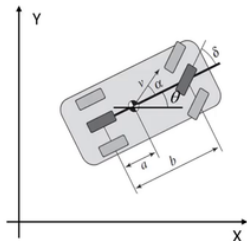
Observability matrix:

$$W_o = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Compute the determinant:

$$\det(W_o) = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0 \cdot 0 - 1 \cdot 0 = 0$$

The system is not observable!



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So, look at our vehicle steering example with heading angle as output:

$$\begin{bmatrix} C \\ CA \end{bmatrix} x^* = 0 \quad \Rightarrow \quad \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x^* = 0 \quad \Rightarrow \quad x^* = \begin{bmatrix} x_1(0) \neq 0 \\ 0 \end{bmatrix}$$

- Karl J. Astrom and Richard M. Murray *Feedback Systems*. Version v3.0i. Princeton University Press. September 2018. Chapter 8.